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Creating an Application in SalesForce.com using Apex Programming Language

Apex is a **strongly-typed, object-oriented programming language** developed by **Salesforce**. It is used to execute **backend operations** on the **Salesforce platform**. Apex is similar to Java and **runs on Salesforce servers**, allowing developers to create custom functionalities like:

- Automating business processes
- Implementing triggers and workflows
- Managing databases with SOQL (Salesforce Object Query Language)
- Integrating with external systems

Scenario	Example
Triggers	Auto-update fields when a record is inserted/updated
Batch Processing	Process thousands of records in chunks
Web Services & Integrations	Call external APIs and handle responses
Scheduled Jobs	Run a task every night (e.g., email reminders)
Security & Access Control	Restrict data access using Apex Sharing Rules

Reference Link for depth knowledge:

https://developer.salesforce.com/docs/atlas.en-us.apexcode.meta/apexcode/apex_dev_guide.htm

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Step 1: signup here

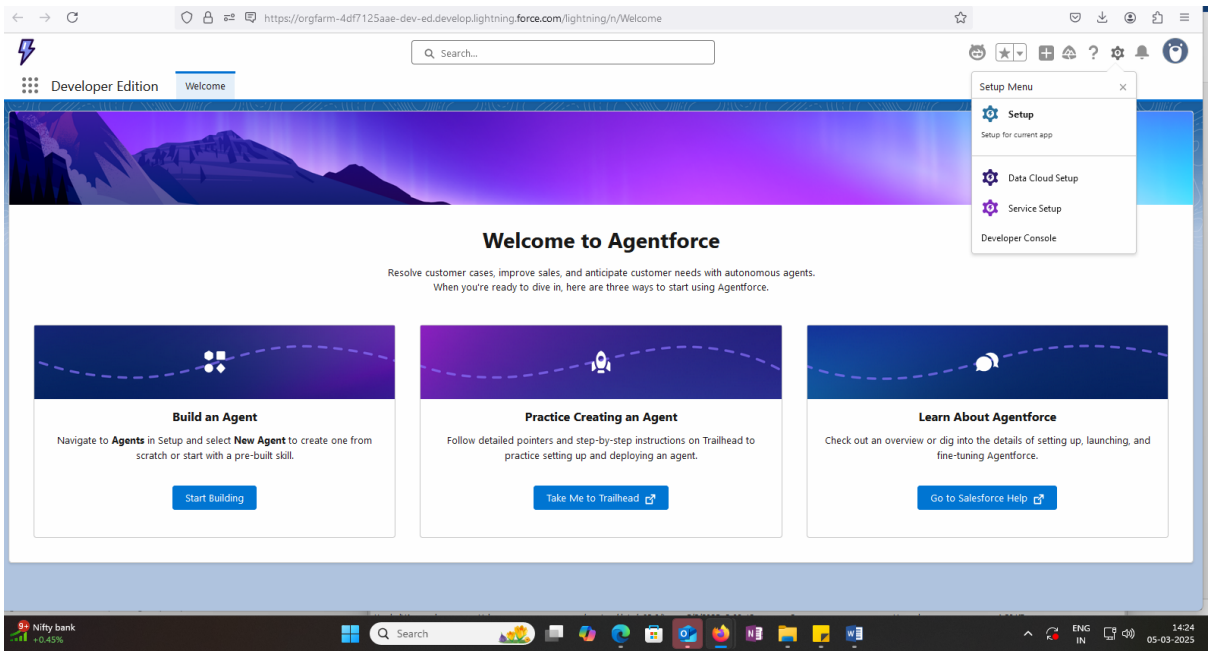
https://www.salesforce.com/form/developer-signup/?d=pb

Step 2: login here

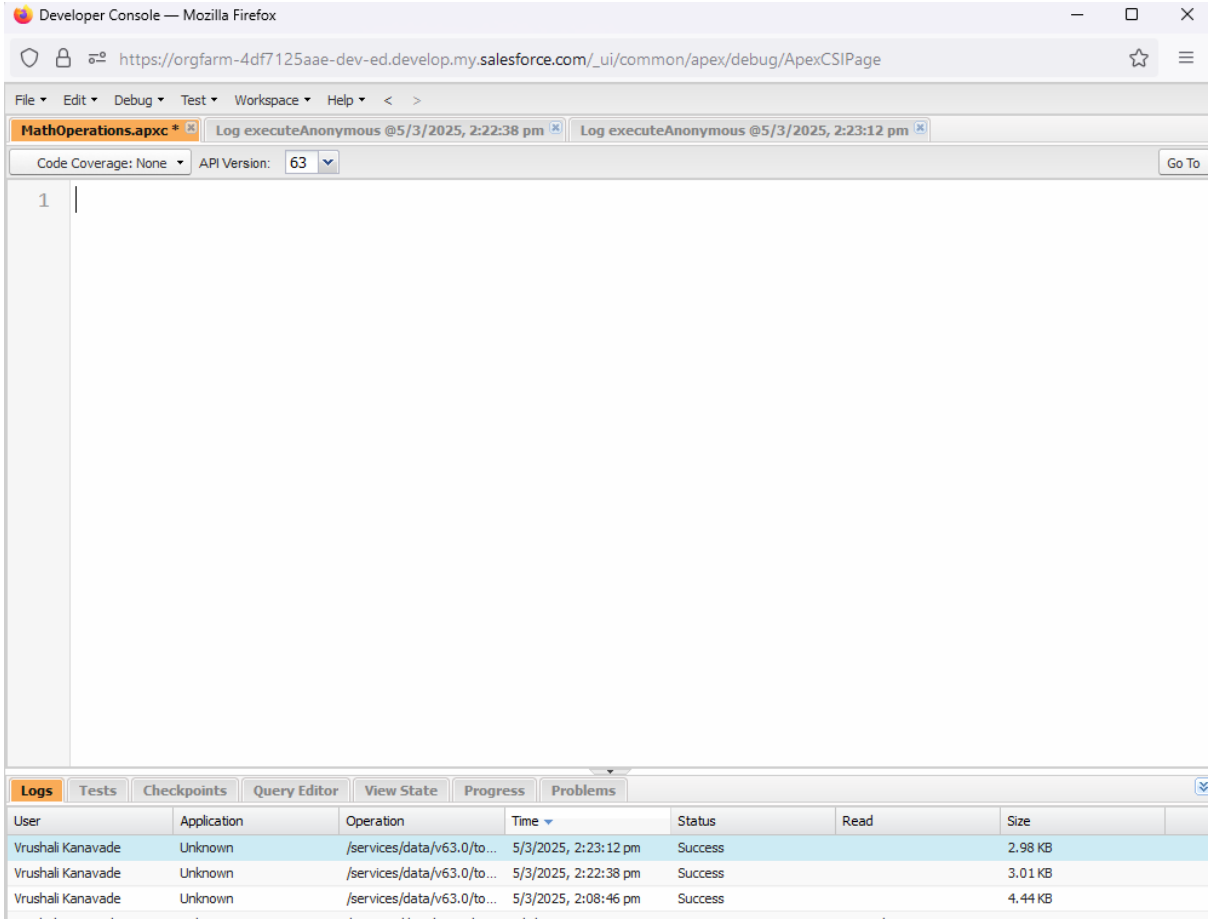
https://login.salesforce.com/

check your email for easy access of console where you get login link as well

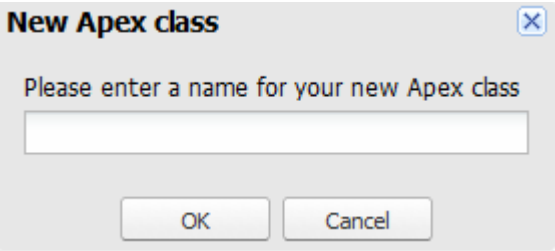
Click on developer console: in right most corner



You will see below console window :



Fo to file -> select new -> select apex class

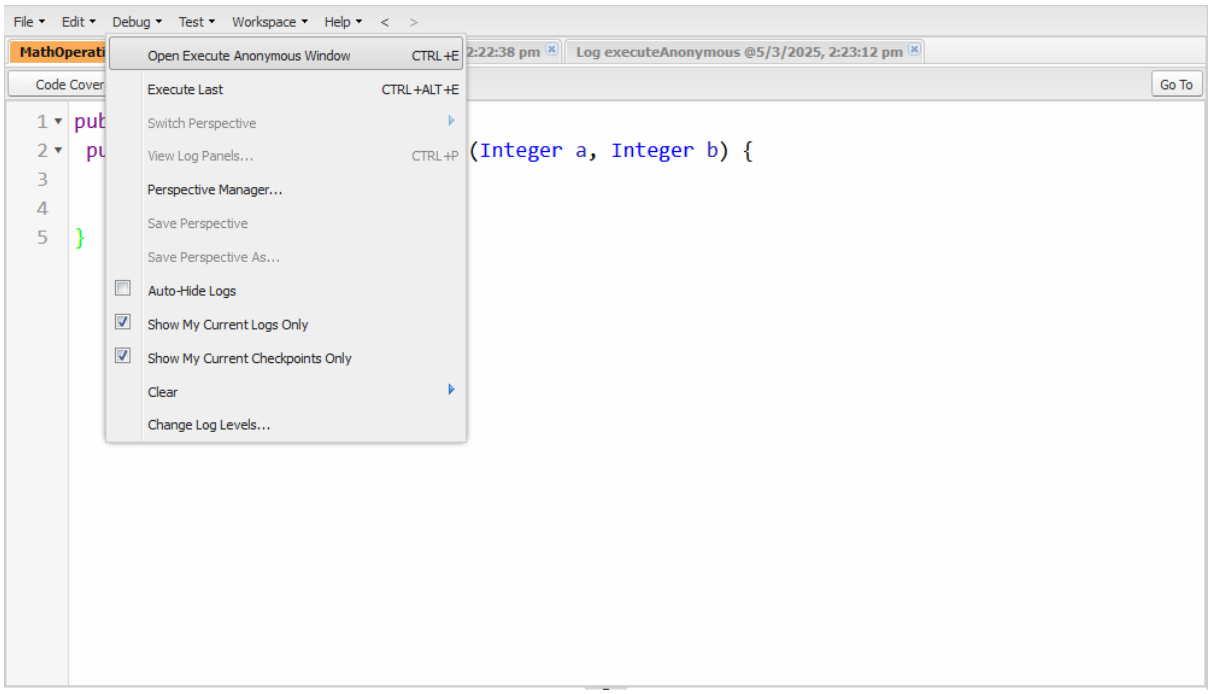


Write class name as per your requirement for example

Code : A method that takes two numbers as input and returns their sum

```
public class MathOperations {
    public static Integer addNumbers(Integer a, Integer b) {
        return a + b;
    }
}
```

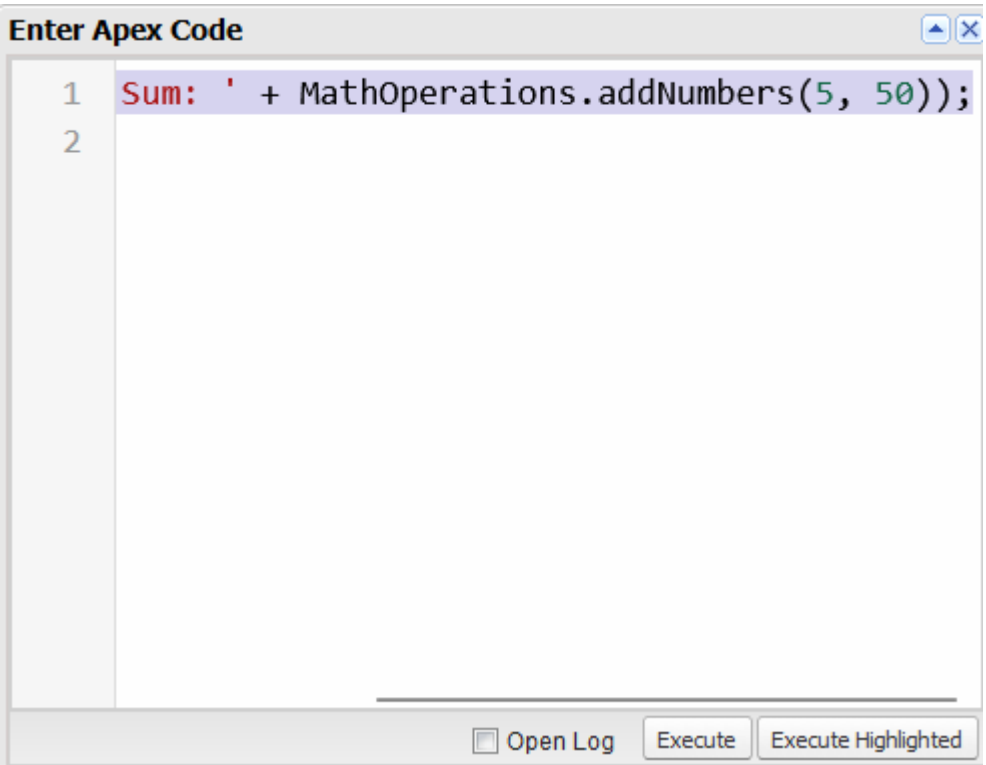
So here class name will be “ MathOperations”



Then go to debug -> Open Execute Anonymous Window ->

For example : for above code now I will give

System.debug('Sum: ' + MathOperations.addNumbers(5, 50));



Then Execute

File Edit Debug Test Workspace Help < >

.apxc * | Log executeAnonymous @5/3/2025, 2:22:38 pm | Log executeAnonymous @5/3/2025, 2:23:12 pm | Log executeAnonymous @5/3/2025, 2:46:46 pm

Execution Log

Timestamp	Event	Details
14:46:46:001	USER_INFO	[EXTERNAL]|005gL000000NMsf|vmkanavade537@agentforce.com|(GMT-08:00) Pacific Standard Time (America/Los_Angeles)|GMT-08:00
14:46:46:001	EXECUTION_ST...	
14:46:46:001	CODE_UNIT_ST...	[EXTERNAL]|execute_anonymous_apex
14:46:46:001	HEAP_ALLOCATE	[93]|Bytes:3
14:46:46:001	HEAP_ALLOCATE	[98]|Bytes:152
14:46:46:001	HEAP_ALLOCATE	[415]|Bytes:408
14:46:46:001	HEAP_ALLOCATE	[428]|Bytes:408
14:46:46:001	HEAP_ALLOCATE	[536]|Bytes:48
14:46:46:001	HEAP_ALLOCATE	[247]|Bytes:6
14:46:46:001	HEAP_ALLOCATE	[EXTERNAL]|Bytes:1
14:46:46:001	STATEMENT_EX...	[1]|
14:46:46:001	STATEMENT_EX...	[1]|
14:46:46:001	HEAP_ALLOCATE	[1]|Bytes:5
14:46:46:001	HEAP_ALLOCATE	[68]|Bytes:5
14:46:46:001	HEAP_ALLOCATE	[74]|Bytes:5
14:46:46:001	HEAP_ALLOCATE	[82]|Bytes:7
14:46:46:001	SYSTEM_MODE_...	false
14:46:46:001	HEAP_ALLOCATE	[1]|Bytes:5
14:46:46:007	HEAP_ALLOCATE	[1]|Bytes:1
14:46:46:007	METHOD_ENTRY	[1]|01pgL000000H6FV|MathOperations.MathOperations()
14:46:46:007	STATEMENT_EX...	[1]|

This Frame | Executable | Debug Only | Filter | Click here to filter the log

Logs | Tests | Checkpoints | Query Editor | View State | Progress | Problems

User	Application	Operation	Time	Status	Read	Size
Vrushali Kanavade	Unknown	/services/data/v63.0/to...	5/3/2025, 2:46:46 pm	Success		2.98 KB
Vrushali Kanavade	Unknown	/services/data/v63.0/to...	5/3/2025, 2:23:12 pm	Success		2.98 KB
Vrushali Kanavade	Unknown	/services/data/v63.0/to...	5/3/2025, 2:22:38 pm	Success		3.01 KB
Vrushali Kanavade	Unknown	/services/data/v63.0/to...	5/3/2025, 2:08:46 pm	Success		4.44 KB
Vrushali Kanavade	Unknown	/services/data/v63.0/to...	5/3/2025, 2:08:12 pm	Success	Unread	4.39 KB
Vrushali Kanavade	Unknown	/services/data/v63.0/to...	5/3/2025, 2:07:43 pm	Success		4.4 KB

Filter | Click here to filter the log list

Select Debug only

And check output

Developer Console — Mozilla Firefox

https://orgfarm-4df7125aae-dev-ed.develop.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

File Edit Debug Test Workspace Help < >

apxc * | Log executeAnonymous @5/3/2025, 2:22:38 pm | Log executeAnonymous @5/3/2025, 2:23:12 pm | Log executeAnonymous @5/3/2025, 2:46:46 pm

Execution Log

Timestamp	Event	Details
14:46:46:008	USER_DEBUG	[1]|DEBUG|Sum: 55

This Frame | Executable | Debug Only | Filter | Click here to filter the log

Logs | Tests | Checkpoints | Query Editor | View State | Progress | Problems

User	Application	Operation	Time	Status	Read	Size
Vrushali Kanavade	Unknown	/services/data/v63.0/to...	5/3/2025, 2:46:46 pm	Success		2.98 KB
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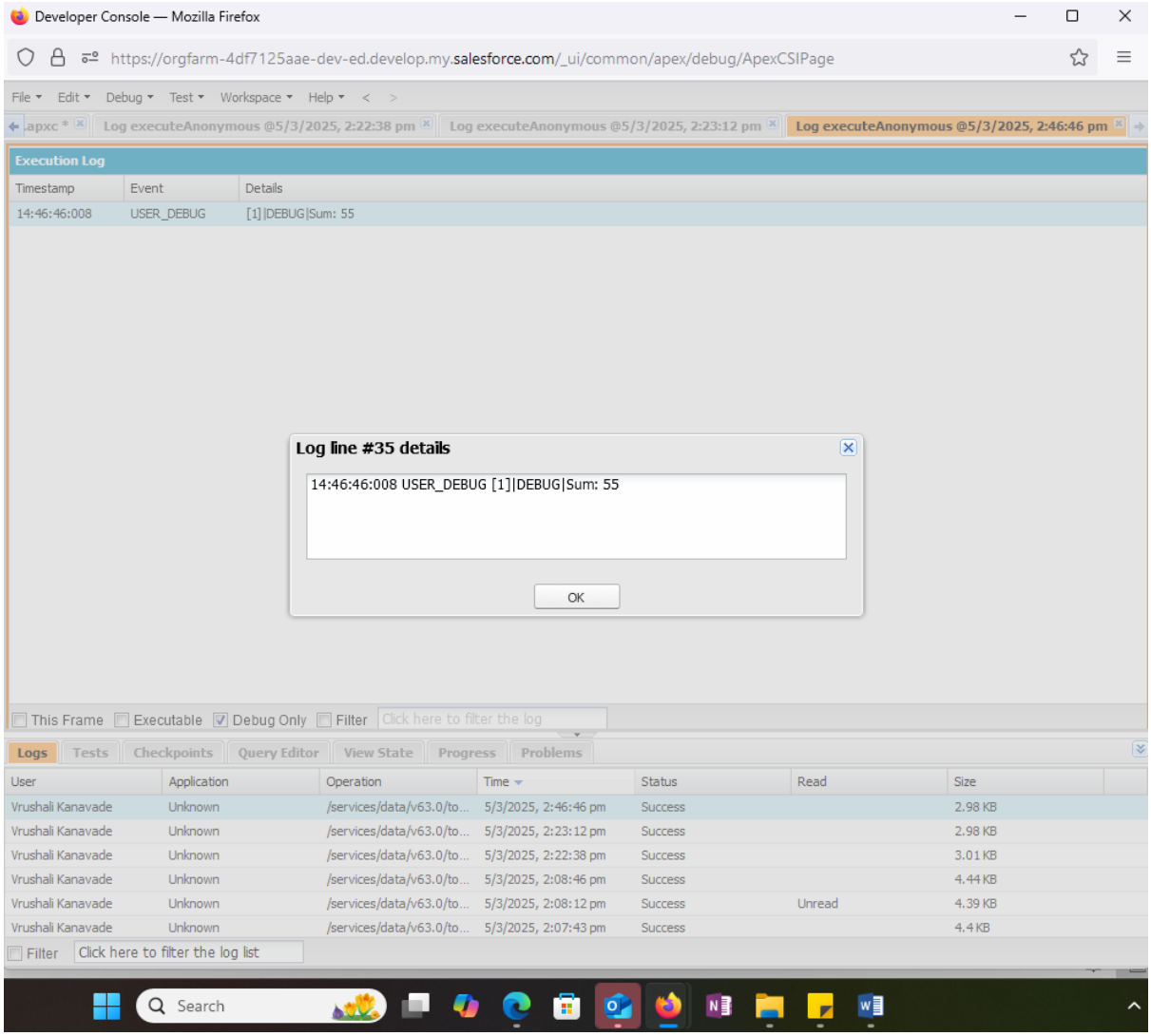
Filter | Click here to filter the log list


Developer Console — Mozilla Firefox
https://orgfarm-4df7125aae-dev-ed.develop.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage
File Edit Debug Test Workspace Help
Log executeAnonymous @5/3/2025, 2:22:38 pm
Log executeAnonymous @5/3/2025, 2:23:12 pm
Log executeAnonymous @5/3/2025, 2:46:46 pm
Execution Log
Timestamp Event Details
14:46:46:008 USER_DEBUG [1]|DEBUG|Sum: 55
Log line #35 details
14:46:46:008 USER_DEBUG [1]|DEBUG|Sum: 55
OK
This Frame Executable Debug Only Filter Click here to filter the log
Logs Tests Checkpoints Query Editor View State Progress Problems
User Application Operation Time Status Read Size
Vrushali Kanavade Unknown /services/data/v63.0/to... 5/3/2025, 2:46:46 pm Success 2.98 KB
Vrushali Kanavade Unknown /services/data/v63.0/to... 5/3/2025, 2:23:12 pm Success 2.98 KB
Vrushali Kanavade Unknown /services/data/v63.0/to... 5/3/2025, 2:22:38 pm Success 3.01 KB
Vrushali Kanavade Unknown /services/data/v63.0/to... 5/3/2025, 2:08:46 pm Success 4.44 KB
Vrushali Kanavade Unknown /services/data/v63.0/to... 5/3/2025, 2:08:12 pm Success Unread 4.39 KB
Vrushali Kanavade Unknown /services/data/v63.0/to... 5/3/2025, 2:07:43 pm Success 4.4 KB
Filter Click here to filter the log list
Search